

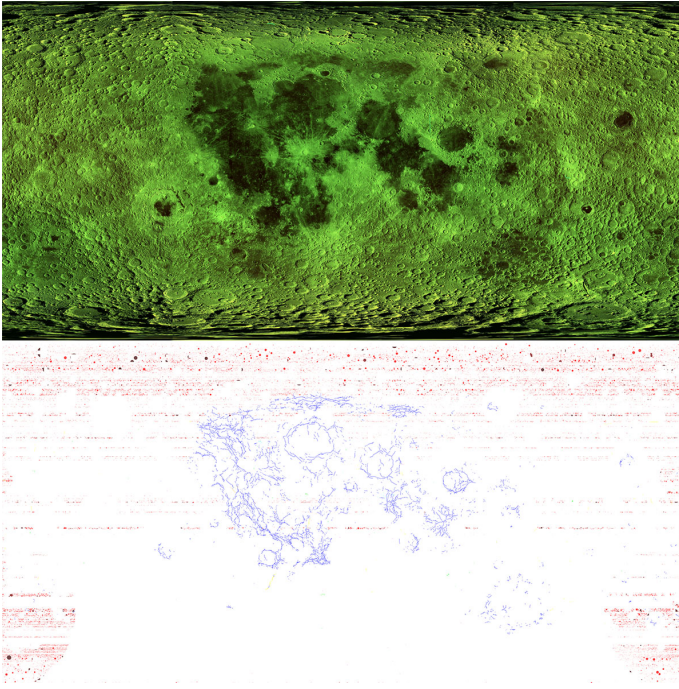


Fig. 2. Color-coded mapping scheme for the seven classified lunar surface features: 0 – Impact Crater (Red), 1 – Pit Skylight (Green), 2 – Wrinkle Ridge (Blue), 3 – Lobate Scarp (Yellow), 4 – Irregular Mare Patch (Cyan), 5 – Apollo Site (Magenta), and 6 – Candidate Rille (Orange).

3. YOLO-Based Detection of Lunar Surface Features

You Only Look Once (YOLO) eliminates the region proposal stage entirely. The input image is divided into a grid, and each grid cell simultaneously predicts both the bounding box coordinates and the class probabilities of any object it contains. All predictions are generated in a single forward pass through the network — the image enters, and detections are produced immediately. This architecture makes YOLO significantly faster than R-CNN-based methods while maintaining competitive accuracy, making it well-suited for real-time detection tasks.

3.1. Architecture

The object detection framework in this study is built entirely upon the YOLOv8 architecture. To optimize feature extraction, lunar surface features are categorized into two primary groups based on their morphological characteristics (as illustrated in Fig. 3):

- **Group f_1 (Two-Dimensional Geometry):** Features characterized by definitive areal structures (e.g., circular or oval geometries), including Craters, Pit Skylights, Irregular Mare Patches, and Apollo Sites.
- **Group f_2 (One-Dimensional Geometry):** Linear and curvilinear features, including Wrinkle Ridges, Lobate Scarps, and Candidate Rilles.

A significant challenge within our dataset is a severe class imbalance, which is illustrated in Fig. 2. Training a single model across all classes simultaneously would lead to poor generalization and high false-negative rates for underrepresented features. To mitigate this effect, we implement a decoupled, multi-model

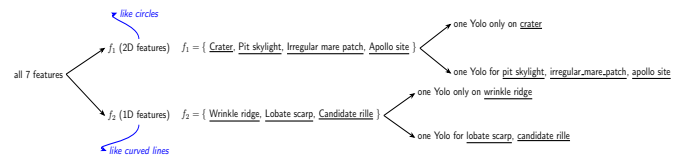


Fig. 3. The division of lunar surface features based on geometric morphology.

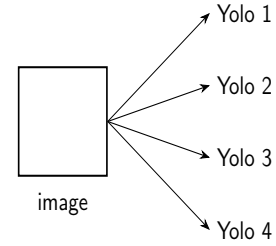


Fig. 4. Implementation and workflow of the four decoupled YOLOv8 models operating as an ensemble framework. Each model is trained on a specific morphological feature group to mitigate severe class imbalance across the dataset.

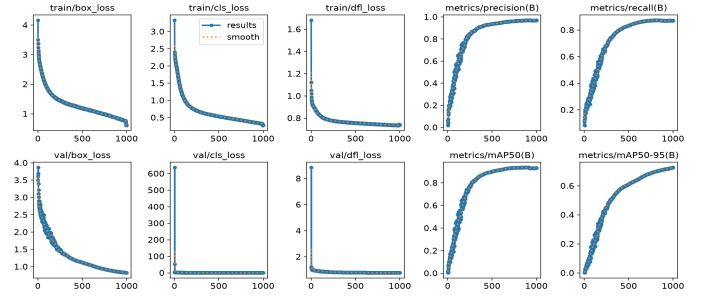


Fig. 5. Training process of the YOLO model focused exclusively on crater detection.

ensemble approach rather than training a single monolithic network for each macro-group.

Consequently, individual YOLOv8 models are dedicated to specific subsets:

- Within the f_1 group, one YOLOv8 model is trained exclusively on Craters, while a separate model is trained to detect Pit Skylights, Irregular Mare Patches, and Apollo Sites.
- Within the f_2 group, the workload is similarly divided: one model is dedicated solely to Wrinkle Ridges, and another handles Lobate Scarps and Candidate Rilles.

This modular architecture ensures that high-frequency features (like craters) do not overwhelm the gradient updates necessary for detecting rarer lunar structures Fig. 4.

3.2. Model Evaluation

The decoupled YOLOv8 models were evaluated using the mean Average Precision at an IoU threshold of 0.50 (mAP50). The models targeting prominent features achieved high accuracy: both the Crater model and the Wrinkle Ridge model reached an mAP50 of 93%.

Conversely, rarer and geometrically complex features proved more challenging. The model for Irregular Mare Patches and

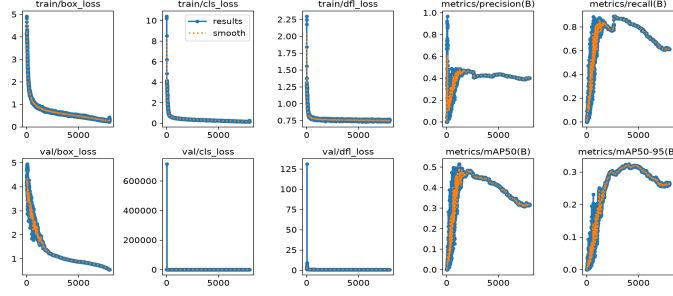


Fig. 6. Training process of the YOLO model focused exclusively on pit skylights, irregular mare patches, and Apollo sites.

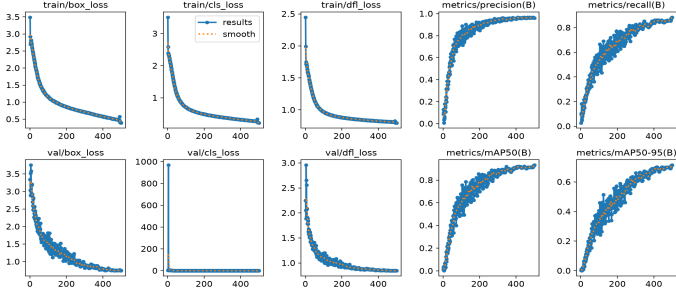


Fig. 7. Training process of the YOLO model focused exclusively on wrinkle ridges.

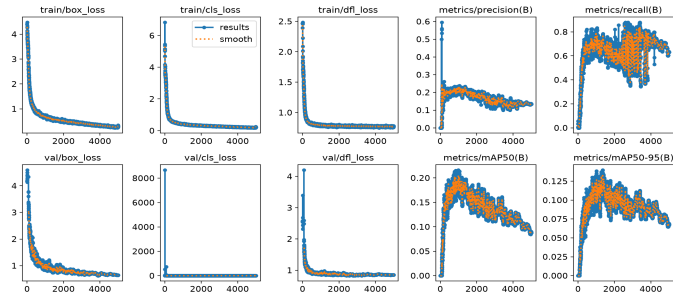


Fig. 8. Training process of the YOLO model focused exclusively on lobate scarp and candidate rilles.

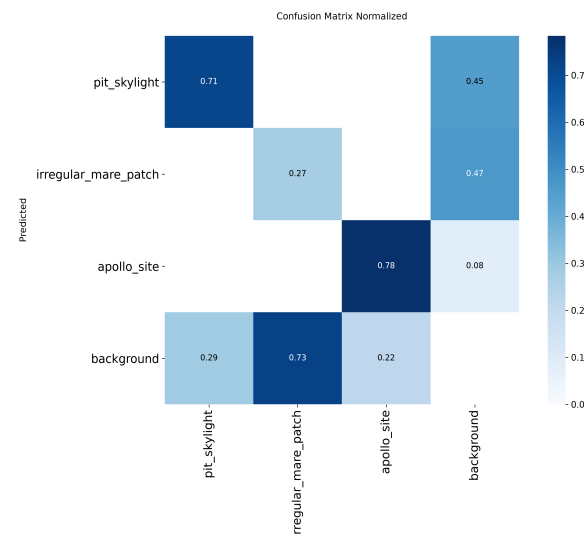


Fig. 9. Confusion matrix evaluated at an IoU threshold of 0.50 for the specialized f_1 subgroup model. The strong diagonal trend demonstrates high classification accuracy with minimal cross-class misclassification among the rare lunar features.

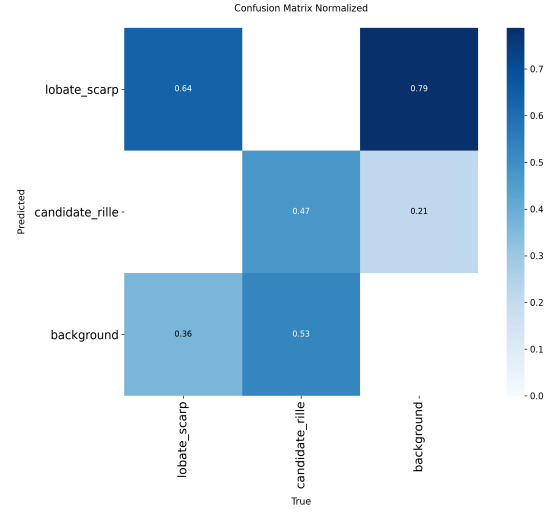


Fig. 10. Confusion matrix evaluated for the specialized f_2 subgroup model. The strong diagonal trend demonstrates high classification accuracy and robust discriminative performance with minimal cross-class misclassification among the designated lunar features.

Apollo Sites achieved an mAP50 of only 50%, while the model for Lobate Scarps and Candidate Rilles yielded an mAP50 of just 25%.

All models were trained using a T4 GPU with 16GB of VRAM.

Figures 5, 6, 7, and 8 show the training process for each model, while Figs. 9 and 10 show the confusion matrices of the two YOLO models that recognize more than one class.

3.3. Validating the Models

Before using the YOLO models to predict lunar south pole features, we must first answer a key question: do the YOLO models produce validated predictions?

One of the powerful advantages of YOLO is its ability to recognize each feature's location and class along with an associated probability, since YOLO performs probabilistic instance segmentation. This architecture allows us to compute the probability distribution of each feature across the lunar surface. By utilizing these results, we can then validate whether our model performs well.

3.3.1. Why YOLO

However, why is YOLO used to determine the probability distributions of features rather than other architectures?

Mask R-CNN is similar to YOLO in that it also performs instance segmentation, producing a mask, classification, and bounding box for each detected object. At first glance, this level of detail might seem more suitable than YOLO's output. However, Mask R-CNN is not better suited than YOLO for computing probability distributions of features, for two main reasons.

First, Mask R-CNN is slower and second, it provides more detailed information about each feature than is necessary for this task. When the goal is simply to calculate probability distributions at a large scale, fine-grained details—such as the exact shape or mask of a feature—are not important.

The other model, U-Net, which performs semantic segmentation, is unsuitable for this purpose because semantic segmenta-

tion cannot distinguish between two instances of the same class, particularly when they are close to one another. Since the number of individual features matters for computing probability distributions—especially when features are spatially close—semantic segmentation is not an appropriate approach for this task.

3.4. Extracting Information

Based on the trained YOLO architecture, I predicted each lunar surface feature with a confidence greater than 50%. I then calculated the probability distribution of each feature across the entire lunar surface and, for each probability density, computed its 2D Fourier transform (Fig. 11).

At first glance, only the Irregular Mare Patch and Candidate Rille features appear to exhibit interesting Fourier transforms; the transforms of the other features resemble white noise, suggesting that these features are randomly distributed with no meaningful spatial structure. However, this is a premature conclusion.

To extract as much information as possible and achieve greater accuracy, I applied Singular Value Decomposition (SVD) to each Fourier transform (Fig. 12). This analysis yields extremely interesting results.

3.4.1. Impact Crater

As shown in Fig. 12 (1st row, 1st column), craters are locally distributed in a horizontal, linear pattern across the lunar surface, which produces a vertical bright line in the corresponding Fourier transform. In general, any directional pattern observed in the original image appears perpendicular to that direction in the Fourier transform.

3.4.2. Candidate Rille

Linear and arcuate rilles are parallel-walled, graben-like features that result from horizontal crustal tension (Solomon, S. C. 1980). The first reconstructed 2D FFT (Fig. 12, last row, third column) resembles white noise, suggesting that Candidate Rilles are, overall, randomly distributed across the lunar surface.

However, the second reconstruction (last row, fourth column) shows that the recurrence frequency of Candidate Rilles across the lunar surface is discrete, occurring only at specific frequencies. Additionally, the fifth and sixth reconstructions reveal a preferred directionality in the 2D Fourier transform, which is consistent with other scientific findings (Solomon, S. C. 1980).

As you can see in the Fig. 13 schematic illustration of subsidence and bending stress due to the load of mare basalt fill in a basin on an elastic lithospheric plate of thickness T . The radial distance to the zone of extension and the magnitudes of both subsidence and stress depend strongly on T . which means that we will have the specific frequencies (Fletcher, Raymond 1974). Additionally, Fig. 14 shows that rilles tend to be either centroidal (radiating from a central point) or linearly parallel, which explains the preferred directionality observed in the FFT.

3.4.3. Irregular Mare Patch

Irregular Mare Patches (IMPs) are small topographic anomalies, ranging from 100 to 5,000 meters in size, located on the lunar nearside maria. They consist of two distinct units: a topographically lower, rough/blocky uneven deposit, and a stratigraphically

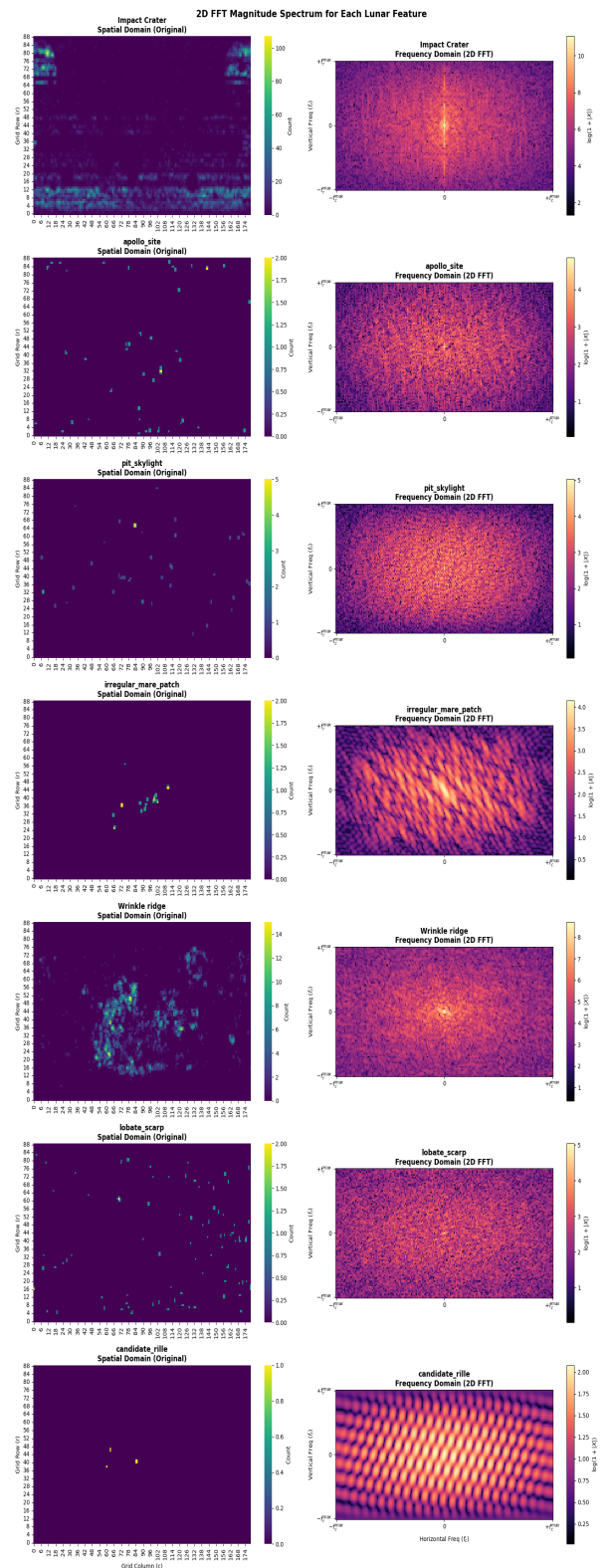


Fig. 11. Probability distributions of each feature over the entire lunar surface (left column) and their corresponding 2D Fourier transforms (right column).

ally higher, younger smooth deposit with steep, lobate margins (Braden, S. 2014).

The first reconstruction (Fig. 12, 4th row, 3rd column) shows white noise, consistent with IMPs occurring as isolated areas

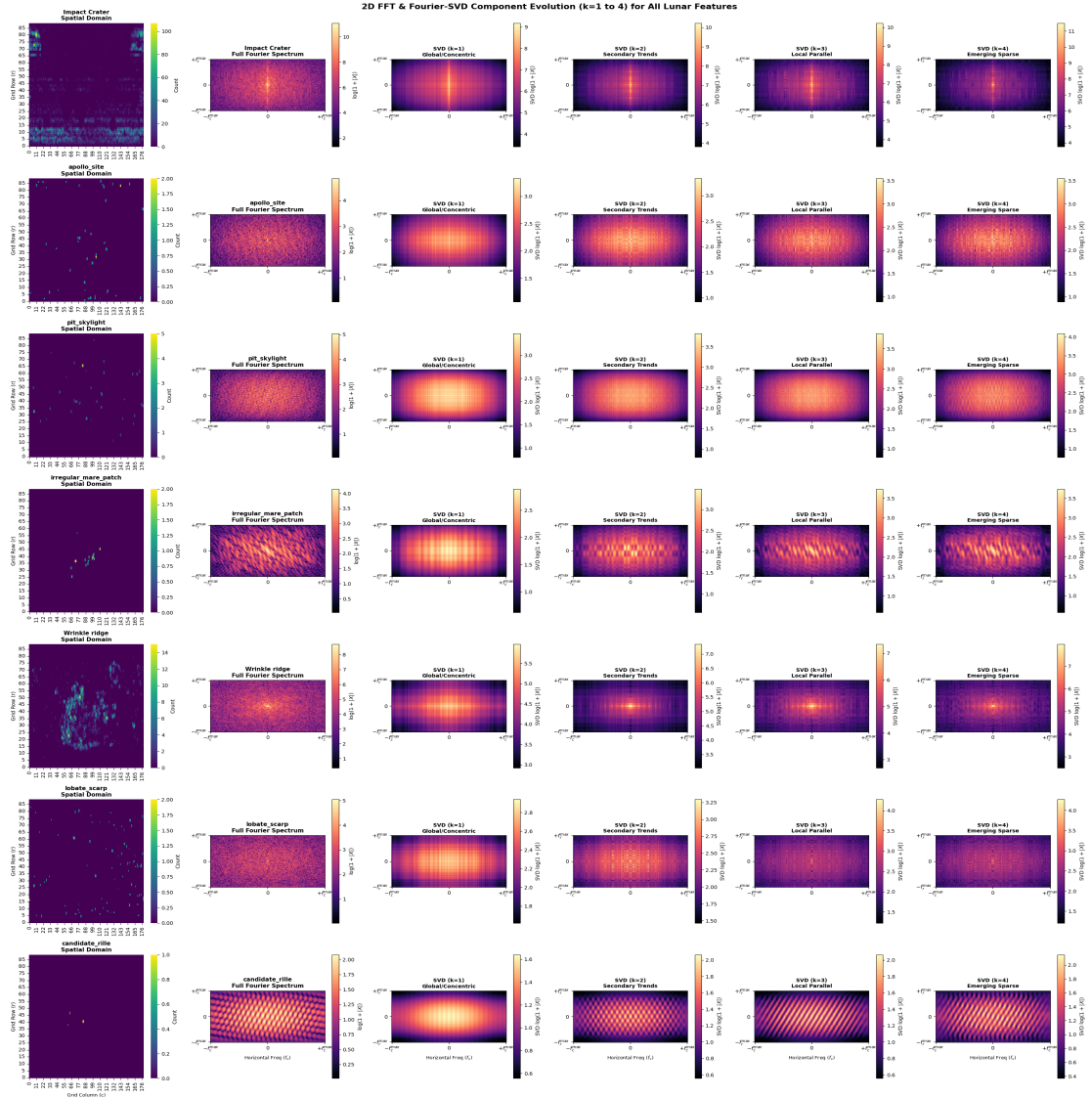


Fig. 12. Column 1 shows the probability distribution; column 2 shows the corresponding 2D Fourier transform. Columns 3, 4, and 5 show the reconstructed 2D Fourier transform using the largest, second-largest, and third-largest eigenvalues, respectively.

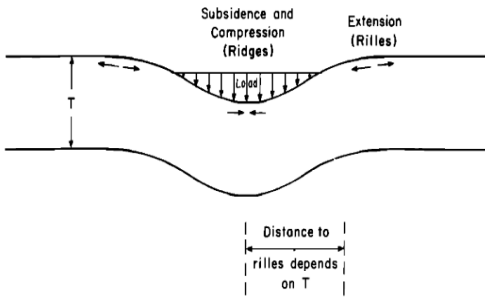


Fig. 13.

on the lunar surface, entirely surrounded by uneven deposits. Smaller IMPs, however, often have smooth deposits connected to the surrounding mare rather than fully isolated—reflected as preferred direction of frequencies in the 4th row, 5th and 6th columns (Fig. 12). The smooth deposits consistently show lobate margins and steep boundary slopes, superposing the uneven

deposits (Fig. 15), which corresponds to the specific frequency seen in the 4th row, 4th column (Fig. 12) (Braden, S. 2014).

3.4.4. Wrinkle Ridge

These features are low, sinuous ridges formed on the mare surface that can extend for several hundred kilometers. Wrinkle ridges are tectonic features created after the lava cooled and solidified. Yue, Z. (2015) categorizes wrinkle ridges into three groups: concentric, parallel, and isolated ridges (Fig. 16). As a result, this behavior is visible in the 5th row, 3rd column (Fig. 12), where the horizontal preferred direction corresponds to parallel and concentric wrinkle ridges. The other reconstructions resemble the preferred direction of white noise, due to isolated wrinkle ridges. Additionally, wrinkle ridges are generally separated vertically (Fig. 17).

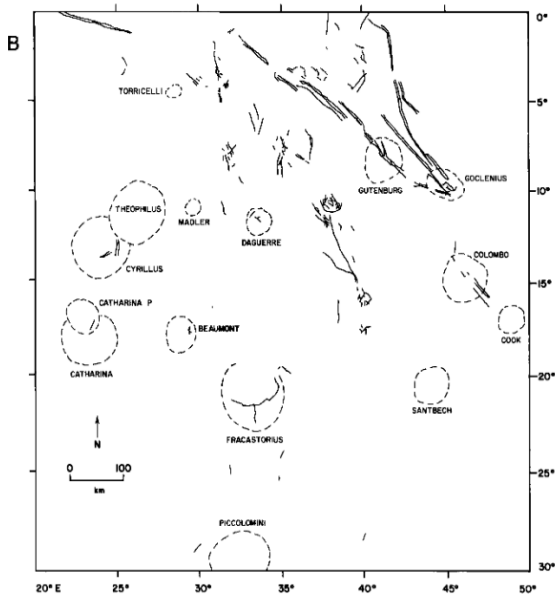


Fig. 14. Shows how Candidate rille distributed.

3.4.5. Lobate Scarp

Lobate scarps are common tectonic landforms on terrestrial planets, representing the surface manifestation of thrust faults formed by contraction of the lunar surface J. D. Clark (2016).

As shown in Fig. 18, lobate scarps are distributed in isolation; however, since some form in linear succession, their first FFT reconstruction shows a preferred direction, followed by reconstructions resembling white noise, as seen in the 6th row of Fig. 12.

3.4.6. Appollo site And Pit Skylight

As shown in Fig. 12, they appear to be distributed completely randomly across the lunar surface. Finally, we can conclude our analysis of Fig. 12 by examining Fig. 19.

3.5. South Pole Evaluation

To evaluate the generalization capability of the ensemble, the four trained YOLOv8 models were deployed to predict lunar surface features across the lunar South Pole dataset. Utilizing a confidence threshold of $\geq 50\%$, the predictive breakdown revealed a heavy dominance of crater impacts: Craters accounted for 98.0% of the detected features, followed by Wrinkle Ridges at 1.4%, and Lobate Scarps at 0.6%. These distributions are visually detailed in the classification mapping presented in Fig. 20.

4. Conclusions

1. Decoupled ensemble works well for imbalanced classes — Splitting detection across four specialized YOLOv8 models (by 2D vs. 1D morphology) effectively handled the severe class imbalance, with high-frequency features like Craters and Wrinkle Ridges reaching 93
2. Fourier + SVD analysis on Yolo predictions reveals real spatial structure — Even features that initially looked randomly distributed. It showed meaningful directional/frequency patterns.

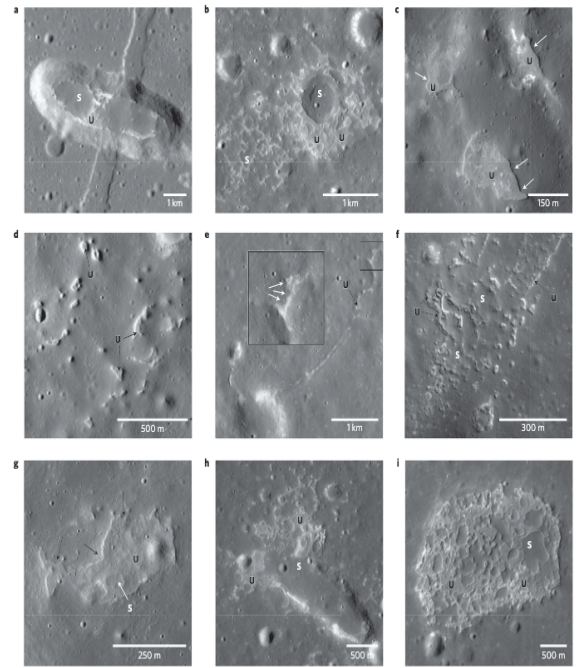


Fig. 15. Examples of smooth and uneven deposits marked with 'S' and 'U'. a, Topographic depression containing the Sosigenes IMP. b, Maske-lyne IMP. Circular topographic high right of centre. c, Manilus-1 IMP. Sharp, lobate contacts (white arrows) where the smooth deposits superposed the uneven deposit. d, IMP inside Hyginus crater. e, Carrel-1 IMP. Narrow, discontinuous sections of uneven deposits may be a vent rim remnant. The inset highlights three lobate contacts (white arrows). f, IMP north of Aristarchus crater. g, Maclear-1 IMP. Black arrows point to lobate margins. h, Cauchy-5 IMP. Superposed on a 6 km diameter volcano. i, Ina exhibits numerous connected and isolated smooth deposits.

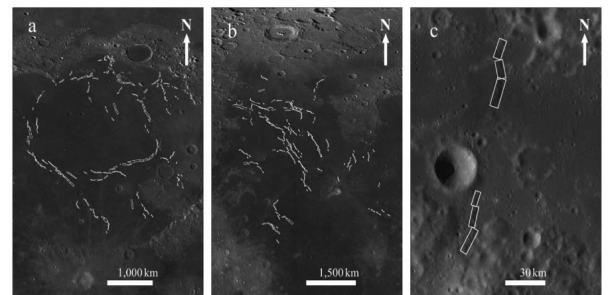


Fig. 16. Classification and grouping of the extracted wrinkle ridges.

3. After confirming that the YOLO models produce meaningful and validated predictions, they were used to predict lunar surface features at the South Pole. The presence of Wrinkle Ridges and Lobate Scarps in this region is consistent with expectations, since both features are tectonic in origin and result from contraction of the lunar surface after cooling. Given that the South Pole is a colder region, it is plausible that such contraction-related features would be more prevalent there but as open question we can say the same for pit skylight but Yolo models could not recognize in the south pole.

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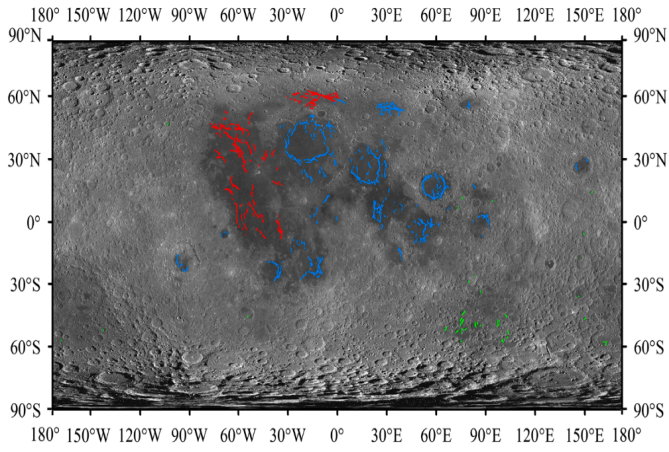


Fig. 17. Global distribution of the extracted lunar wrinkle ridges. In which the blue, red, and green lines represent the extracted segments of concentric, parallel, and isolated ridges, respectively.



Fig. 18. Distributed of Lobate Scarp.

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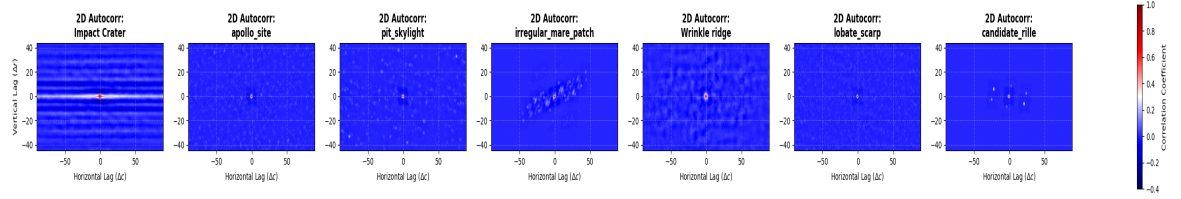


Fig. 19. Self-correlation of each feature.

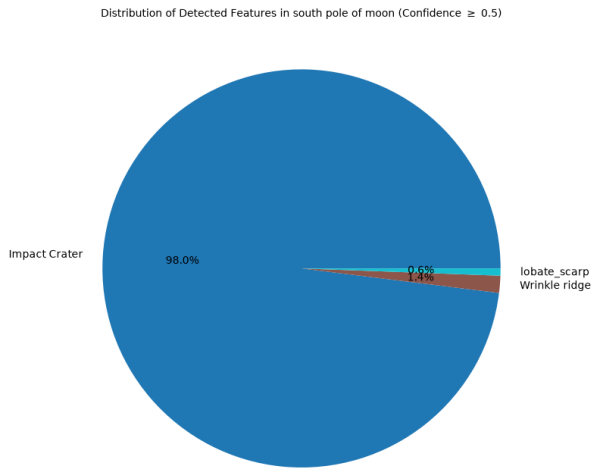


Fig. 20. Percentage distribution of the predicted lunar surface features in south pole.

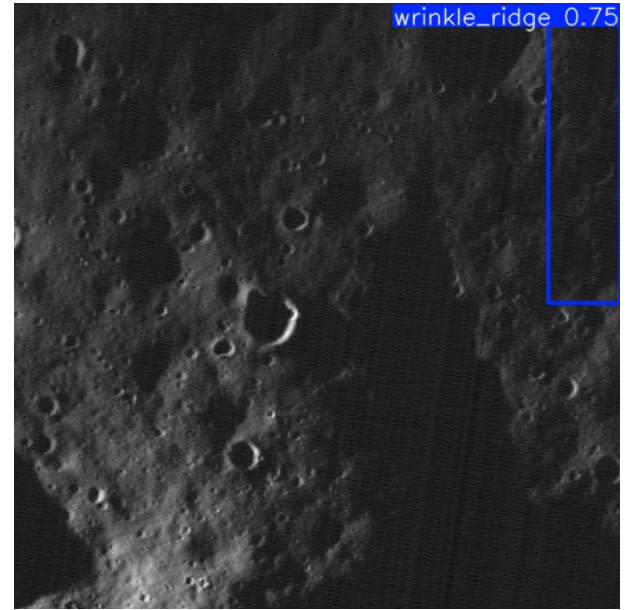


Fig. 22. tile₀00512.

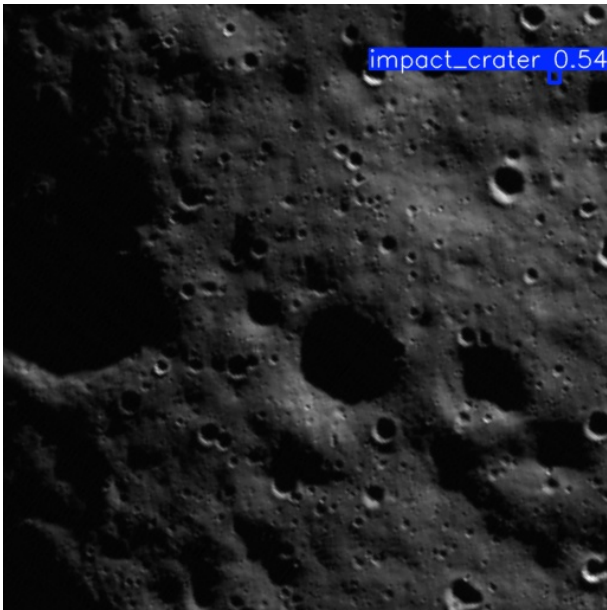


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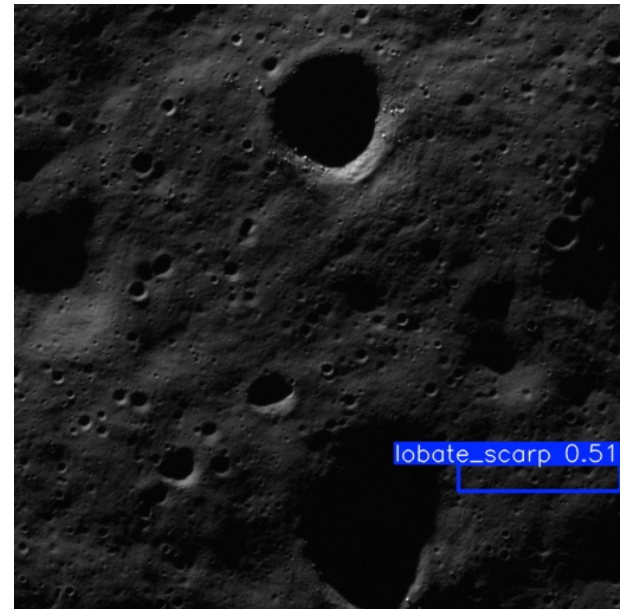


Fig. 23. tile₀01395.